Lecture 7: General Autoregressive Models

Ignasi Cos

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1 Introduction

The purpose of this lecture is to present the general AR(p) model, after having seen the AR(1) case in the previous lecture.

In all the lecture $Z = \{Z_j,\ j \in \mathbb{Z}\}$ will be a centered white noise with variance $\sigma^2 > 0$, and $\{Y_j, j \in \mathbb{Z}\}$ will be a time series. Note that for simplicity in the presentation we consider time series on $\mathbb{Z}$. In practice we have $j \geq 0$ or $j \geq 1$. Recall the operator B introduced in Lecture 1 that acts as $BY_j := Y_{j-1}$.

We can write the AR(1) equation as

$$Y_j = \phi B Y_j + Z_j$$

or

$$(\mathrm{Id} - \phi B) Y_j = Z_j.$$

Therefore, if $(\mathrm{Id} - \phi B)^{-1}$ exists, we can write

$$Y_j = (\mathrm{Id} - \phi B)^{-1} Z_j.$$

But on other hand, using the linear causal expression of Y obtained in the previous lecture, we know that the AR(1) model can be written as

$$Y_j = \sum_{i=0}^{\infty} \phi^i B^i Z_j.$$

Therefore, we can define

$$(\mathrm{Id} - \phi B)^{-1} = \sum_{i=0}^{\infty} \phi^i B^i \tag{1.1}$$

provided $\sum_{i=0}^{\infty} \phi^i B^i Z_j$ is well defined in L^2. Note that as a consequence of Corollary 2.4 in the previous lecture this is equivalent to

$$\sum_{i=0}^{\infty} |\phi|^{2i} < \infty.$$

And it is well known that this convergence is equivalent to $|\phi| < 1$, that is the condition on ϕ that guarantees that the AR(1) model is a stationary model.

Notice that equation (**??**) is a formal version of the equality

$$\sum_{i=0}^{\infty} \phi^i x^i = \frac{1}{1-\phi x}$$

that is true if and only if $|\phi x| < 1$. Equivalently, we can write

$$(1-\phi x) \cdot \sum_{i=0}^{\infty} \phi^i x^i = 1, \quad \forall x \colon |x| < \frac{1}{|\phi|}.$$

2 The AR(p) model

Let $\Phi(x)$ be a polynomial of degree p. We can always write

$$\Phi(x) = 1 - \phi_1 x - \phi_2 x^2 - \cdots - \phi_p x^p \tag{2.2}$$

up to a multiplicative constant, dividing the polynomial by the independent coefficient.

Note that this is an extension to any degree of the polynomial $1 - \phi x$ associated to the AR(1) model.

Definition 2.1 *We say that a polynomial $\Phi(x)$ is invertible if it exist a series $\sum_{i=0}^{\infty} \psi_i x^i$ such that*

$$\Phi(x) \sum_{i=0}^{\infty} \psi_i x^i = 1$$

with

$$\sum_{i=0}^{\infty} \psi_i^2 < \infty.$$

Notice that if Φ is invertible we have

$$Y_j = \Phi(B)^{-1} Z_j = \sum_{i=0}^{\infty} \psi_i B^i Z_j$$

and this guarantees the causal and stationary character of Y.

So, the key question is, when Φ is invertible? We have seen that if $p = 1$, the answer is $|\phi| < 1$. But what is the answer when $p > 1$? The following theorem gives it.

Theorem 2.2 *A polynomial $\Phi_p(z)$, with $z \in \mathbb{C}$, is invertible if and only if all its roots are out of the unit circle, that is, if and only if $\{z \colon \Phi_p(z) = 0\} \subseteq \{z \colon |z| > 1\}$.*

Proof: See [**?**], pages 85-86. ■

Remark 2.3 *If $p = 1$, the previous theorem is clear. We have $\Phi(z) = 1 - \phi z$. In this case,*

$$\{z \colon 1 = \phi z\} = \{z \in \mathbb{C} \colon z = 1/\phi\},$$

and

$$\frac{1}{|\phi|} > 1 \Longleftrightarrow |\phi| < 1.$$

We define an AR(p) model as a stationary process $\{Y_j,\, j \in \mathbb{Z}\}$ that satisfies

$$\Phi(B)Y_j = Z_j, \quad j \in \mathbb{Z}. \tag{2.3}$$

where Φ is an invertible polynomial of degree p.

Note that given any polynomial of degree p we can divide it by the independent term and write it as $c\Phi(x)$ where c is a constant and Φ is of the form (**??**). In that case we can erase the constant c from the left hand side of (**??**) and change the variance of the white noise on the right hand side from σ^2 to $(\frac{\sigma}{c})^2$.

Since our AR(p) model is causal we have immediately that it is centered and its auto-covariance function is

$$\mathbb{C}(Y_j, Y_{j+l}) = \sigma^2 \sum_{r=0}^{\infty} \psi_r \psi_{r+l}.$$

But to identify the coefficients of the causal representation is not so easy.

One way is to do it recursively. Imposing

$$\Phi_p(x) \sum_{i=0}^{\infty} \psi_i x^i = 1$$

we have

$$\begin{aligned}
(1 - \phi_1 x - \phi_2 x^2 - \cdots - \phi_p x^p) \sum_{i=0}^{\infty} \psi_i x^i &= 1 \\
\Longleftrightarrow \sum_{i=0}^{\infty} \psi_i x^i - \phi_1 \sum_{i=0}^{\infty} \psi_i x^{i+1} - \cdots - \phi_p \sum_{i=0}^{\infty} \psi_i x^{i+p} &= 1 \\
\Longleftrightarrow \sum_{i=0}^{\infty} \psi_i x^i - \phi_1 \sum_{i=1}^{\infty} \psi_{i-1} x^i - \cdots - \phi_p \sum_{i=p}^{\infty} \psi_{i-p} x^i &= 1.
\end{aligned}$$

Equating the independent coefficient to 1 and the coefficients of the different powers of x to 0, we have

$$\begin{aligned}
&\psi_0 = 1 \\
&\psi_1 - \psi_0 \phi_1 = 0 \\
&\psi_2 - \psi_1 \phi_1 - \psi_2 \phi_2 = 0 \\
&\cdots\cdots,
\end{aligned}$$

that is, in general,

$$\psi_i = \phi_1 \psi_{i-1} + \cdots + \phi_p \psi_{i-p}, \quad i \geq p$$

So, recursively, $\psi_0 = 1$, $\psi_1 = \psi_0 \phi_1$, $\psi_2 = \psi_1 \phi_1 + \psi_0 \phi_2$ and so on.

Note that from the modeling point of view, given a centered second order stationary model, the only important object is the autocorrelation function. The variance is a constant, easy to estimate, and we cannot distinguish what is the weight of the variance of the white noise in the variance of the model.

To determine directly the autocorrelation function we have the following alternative method. Notice that

$$\begin{aligned}\mathbb{C}(Y_j, Y_{j+l}) &= \mathbb{C}(Y_j, \phi_1 Y_{j+l-1} + \cdots + \phi_p Y_{j+l-p} + Z_{j+l}) \\ &= \phi_1 \gamma(l-1) + \phi_2 \gamma(l-2) + \cdots + \phi_p \gamma(l-p), \quad l > 0,\end{aligned}$$

that is,

$$\gamma(l) = \phi_1 \gamma(l-1) + \cdots + \phi_p \gamma(l-p), \quad l > 0.$$

Dividing by $\gamma(0)$ we obtain

$$\begin{aligned}\rho(l) &= \phi_1 \rho(l-1) + \cdots + \phi_p \rho(l-p), \quad l > 0 \\ \rho(0) &= 1 \\ \rho(-l) &= \rho(l),\end{aligned}$$

that is a solvable system of equations of finite differences and the solution is the autocorrelation function of our AR(p) model. This system is called Yule-Walker equations.

3 Example: the AR(2) model

A AR(2) model is a process that satisfies the equation

$$Y_j - \phi_1 Y_{j-1} - \phi_2 Y_{j-2} = Z_j, \quad j \in \mathbb{Z}$$

with ϕ_1 and ϕ_2 real numbers such that the polynomial

$$1 - \phi_1 x - \phi_2 x^2$$

have all roots out of the unit circle.

As we have seen in the previous section, the autocorrelation function satisfies the following Yule-Walker equations:

$$\begin{aligned}\rho(l) &= \phi_1\, \rho(l-1) + \phi_2\, \rho(l-2), \quad l \in \mathbb{Z} \\ \rho(0) &= 1 \\ \rho(-l) &= \rho(l).\end{aligned}$$

The way to solve this equation, see for example [**?**], consists in studying the roots of the polynomial

$$1 - \phi_1 x - \phi_2 x^2,$$

or equivalently to distinguish the different possible values of the discriminant

$$\Delta := \phi_1^2 + 4\phi_2.$$

1. If $\Delta > 0$, the polynomial has two real roots η_1 and η_2. Then,

$$\rho(l) = c_1 \left(\frac{1}{\eta_1}\right)^l + c_2 \left(\frac{1}{\eta_2}\right)^l.$$

2. If $\Delta = 0$, $\phi_1^2 = -4\phi_2$, and we have a unique double root η. In this case,

$$\rho(l) = c_1 \left(\frac{1}{\eta}\right)^l + c_2\, l \left(\frac{1}{\eta}\right)^l .$$

3. Finally, if $\Delta < 0$, we have two conjugated complex roots η_1 and η_2. Then,

$$\rho(l) = c_1 \left(\frac{1}{\eta_1}\right)^l + c_2 \left(\frac{1}{\eta_2}\right)^l = \frac{1}{r^l}\{A_1 \sin(\alpha\, l) + A_2 \cos(\alpha\, l)\}$$

with $r = |\eta_1| = |\eta_2|$.

To determine the constants we can use the so called Yule-Walker equations. We can write the first two difference equations, for $l = 1$ and $l = 2$, and built the system

$$\begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}$$

and so,

$$\begin{aligned} \phi_1 + \rho_1 \phi_2 &= \rho_1 \\ \rho_1 \phi_1 + \phi_2 &= \rho_2 \end{aligned}$$

and isolating ρ_1 and ρ_2 we obtain

$$\begin{aligned} \rho_1 &= \frac{\phi_1}{1 - \phi_2} \\ \rho_2 &= \phi_2 + \frac{\phi_1^2}{1 - \phi_2}. \end{aligned}$$

Using $\rho(0) = 1$ and $\rho(1) = \frac{\phi_1}{1-\phi_2}$ we can determine c_1 and c_2 or A_1 and A_2. This technique can be generalized to AR(p) case writing the first p Yule-Walker equations.

As a particular and concrete example, consider the AR(2) model

$$X_j = 0.7 X_{j-1} - 0.1 X_{j-2} + Z_j,$$

with $Z \sim \mathrm{WN}(0, 1)$.
We have to solve equation

$$1 - 0.7x + 0.1x^2 = 0.$$

The roots are 2 and 5. Then, $\eta_1 = 2$ and $\eta_2 = 5$, that of course are out of the unit circle of $\mathbb{C}$. Then,

$$\rho(l) = c_1 \frac{1}{2^l} + c_2 \frac{1}{5^l}.$$

In particular $\rho(0) = 1$ and so $c_1 + c_2 = 1$. On other hand,

$$\rho(1) = \frac{\phi_1}{1 - \phi_2} = \frac{0.7}{1 + 0.1} = \frac{7}{11}$$

Then, we have to solve the system

$$\begin{aligned}\frac{1}{2}\,c_1 + \frac{1}{5}\,c_2 &= \frac{7}{11}\\ \frac{1}{2}\,c_1 + \frac{1}{2}\,c_2 &= \frac{1}{2},\end{aligned}$$

obtaining

$$\begin{aligned}c_1 &= \frac{16}{11}\\ c_2 &= -\frac{5}{11}.\end{aligned}$$

Therefore,

$$\rho(l) = \frac{1}{11}\left\{16\,\frac{1}{2^l} - 5\,\frac{1}{5^l}\right\}, \quad l \geq 0.$$

Observe that

$$|\rho(l)| \leq \frac{c_1}{2^l} + \frac{c_2}{5^l} \leq \frac{k}{2^l} \overset{l\uparrow\infty}{\longrightarrow} 0.$$

In relation with the determination of ψ_i coefficients we have

$$\begin{aligned}&\psi_0 = 1\\ &\psi_1 = \phi_1 = 0.7\\ &\psi_2 = \phi_1^2 + \phi_2 = 0.7^2 - 0.1 = 0.39\\ &\psi_3 = \phi_1\psi_2 + \phi_2\psi_1 = 0.7 \times 0.39 - 0.1 \times 0.7 = 0.203\\ &\text{etc}\ldots\end{aligned}$$

Note that $|\psi_i| \overset{l\uparrow\infty}{\longrightarrow} 0$ because $\sum_i \psi_i^2 < \infty$.

4 Moving average models (MA)

The moving average model specifies that the output variable is cross-correlated with a non-identical to itself random-variable. Given a centered white noise Z with variance σ^2 and a natural number $q \geq 0$, we define the MA(q) model as the process $\{Y_j,\ j \in \mathbb{Z}\}$ defined as

$$Y_j = Z_j + \theta_1 Z_{j-1} + \cdots + \theta_q Z_{j-q}.$$

where $\theta_1, \ldots, \theta_q$ are real numbers.

The immediate properties of this process are the following:

1. It is a second order process because Z it is.
2. It is a centered process because Z is centered and

$$E(Y_j) = \sum_{l=0}^{q} \theta_l E(Z_{j-l}) = 0.$$

Note that by convention $\theta_0 = 1$.

3. The auto-covariance function is given by

$$\mathbb{C}(Y_j, Y_{j+k}) = \sigma^2 \sum_{i=0}^{q-|k|} \theta_i \theta_{i+|k|}, \quad |k| \leq q$$

and 0 in all the other cases.

Indeed, we can choose $k \geq 0$ without loosing generality, and then,

$$\begin{aligned}
\mathbb{C}(Y_j, Y_{j+k}) &= \mathbb{C}\left(\sum_{r=0}^{q} \theta_r Z_{j-r}, \sum_{s=0}^{q} \theta_s Z_{j+k-s}\right) \\
&= \sum_{r=0}^{q}\sum_{s=0}^{q} \theta_r \theta_s \mathbb{C}(Z_{j-r}, Z_{j+k-s}) \\
&= \sum_{r=0}^{q}\sum_{s=0}^{q} \theta_r \theta_s \sigma^2 \mathbb{1}_{\{j-r=j+k-s\}} \\
&= \sum_{r=0}^{q}\sum_{s=0}^{q} \theta_r \theta_s \sigma^2 \mathbb{1}_{\{s=r+k\}} \\
&= \sum_{r=0}^{q-k} \theta_r \theta_{r+k} \sigma^2.
\end{aligned}$$

Therefore, it is a stationary process because the auto-covariance between Y_j and Y_{j+k} does not depend on j.

4. The variance is

$$\mathbb{V}(Y_j) = \sigma^2 \sum_{r=0}^{q} \theta_r^2, \quad \forall j \in \mathbb{Z}$$

5. The auto-correlation function is given by

$$\rho(k) = \frac{\sum_{r=0}^{q-k} \theta_r \theta_{r+k}}{\sum_{r=0}^{q} \theta_r^2}, \quad 0 \leq k \leq q.$$

and so, it is not null only for $k = 0, 1, \ldots, q$. Notice that this is completely different from the AR case.

In terms of lag operator B we have

$$Y_j = Z_j + \theta_1 B Z_j + \theta_2 B^2 Z_j + \cdots + \theta_q B^q Z_j$$

and putting $B^0 = \mathrm{Id}$ we can write

$$Y_j = \Theta_q(B) Z_j$$

where

$$\Theta_q(x) = 1 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_q x^q$$

is a polynomial of order $q \geq 0$.

Example 4.1 (MA(1) model) *MA(1) model is given by*

$$Y_j = Z_j + \theta Z_{j-1}, \quad j \in \mathbb{Z},\ \theta \in \mathbb{R}.$$

Then, the variance is

$$\mathbb{V}(Y_j) = \sigma^2(1+\theta^2)$$

and the auto-covariance function is null except for lags 0 *and* 1. *In these cases it takes values*

$$\gamma(0) = \sigma^2(1+\theta^2)$$

and

$$\gamma(1) = \sigma^2\theta.$$

Equivalently, $\rho(0) = 1$,

$$\rho(1) = \frac{\theta}{1+\theta^2}$$

and $\rho(l) = 0$ *for any other lag.*

Remark 4.2 *Consider the process*

$$X_j = V_j + \frac{1}{\theta} V_{j-1}$$

with $V \sim WN(0, \theta^2\sigma^2)$. *This is a centered MA(1) process with variance*

$$\left(1 + \frac{1}{\theta^2}\right)\theta^2\sigma^2 = \sigma^2(1+\theta^2)$$

and auto-correlation of order 1 *given by*

$$\rho(1) = \frac{1/\theta}{1+1/\theta^2} = \frac{\theta}{1+\theta^2}.$$

Therefore, an MA(1) model with parameters θ *and* σ^2 *and an MA(1) of parameters* $\frac{1}{\theta}$ *and* $\theta^2\sigma^2$ *have the same structure as second order stationary processes. So, we can assume without loosing generality that* $|\theta| \leq 1$.

5 ARMA models

As the name already suggests, these are models combining autoregressive (AR) and moving average (MA). Typically, they are referred to as ARMA (p,q) models, with p AR terms and q MA terms. In their general form, they may be represented by the following expression:

$$Y_k = \sum_{j=1}^{p} \Phi_j Z_{k-j} + \sum_{i=1}^{q} \Theta_i Y_{k-i}$$

This may also be expressed in terms of the B operator. Let $P_n(x) = 1 - \alpha_1 x - \alpha_2 x - \cdots - \alpha_n x^n$ be a polynomial of degree n with all the roots out of the unit circle, that is, invertible. Then,

$$Y = P(B)Z$$

is a MA(n) model and

$$Y = P(B)^{-1}Z \Longleftrightarrow Z = P(B)Y$$

is an AR(n) model.

Definition 5.1 *Let p and q two fixed natural numbers, $Z \sim WN(0, \sigma^2)$ and $\Phi_p(\cdot)$ and $\Theta_q(\cdot)$ two invertible polynomials of degrees p and q respectively without roots in common. Then, a model ARMA(p,q) is a model that satisfies the equation*

$$\Phi_p(B)Y_j = \Theta_q(B)Z_j, \quad j \in \mathbb{Z}.$$

Remark 5.2 *Using the fact that $\Phi_p(\cdot)$ is invertible we can write*

$$Y_j = \Phi_p(B)^{-1}\Theta_q(B)Z_j = \Theta_q(B)\Phi_p(B)^{-1}Z_j.$$

This fact proves that ARMA models are linear and causal models.

Remark 5.3 *Note that to put positive or negative signs in front of coefficients of polynomial Φ and Θ is irrelevant because coefficients are real numbers. From now on we will consider always polynomials with negative sign in front of all the parameters.*

Example 5.4 (ARMA(1,1) model) *ARMA(1,1) model satisfies*

$$Y_j - \phi\, Y_{j-1} = Z_j - \theta\, Z_{j-1}, \quad j \in \mathbb{Z},$$

with $|\phi| < 1$, $|\theta| < 1$ and $\phi \neq \theta$.

Solving the equation

$$(1 - \theta\, x) = (1 - \phi\, x)\sum_{i=0}^{\infty} \psi_i x^i$$

we can obtain the causal expression. We have

$$\begin{aligned}
1 - \theta\, x &= \sum_{i=0}^{\infty} \psi_i\, x^i - \sum_{i=0}^{\infty} \phi\, \psi_i\, x^{i+1} \\
&= \sum_{i=0}^{\infty} \psi_i\, x^i - \sum_{j=1}^{\infty} \phi\, \psi_{j-1}\, x^j \\
&= \psi_0 + \sum_{j=1}^{\infty} (\psi_j - \phi\, \psi_{j-1}) x^j
\end{aligned}$$

and so,

$$\begin{gathered}
\psi_0 = 1 \\
\psi_1 - \phi\, \psi_0 = -\theta \\
\psi_j = \phi\, \psi_{j-1}, \quad \forall j \geq 2.
\end{gathered}$$

Finally

$$\psi_0 = 1$$
$$\psi_1 = \phi - \theta$$
$$\psi_i = \phi\,\psi_{i-1} = \cdots = \phi^{i-1}\,\psi_1 = \phi^{i-1}(\phi - \theta), \quad i \geq 1.$$

We know that the auto-covariance function is given by

$$\gamma(l) = \sigma^2 \sum_{i=0}^{\infty} \psi_i\,\psi_{i+l}.$$

Then, for $l = 0$ we have

$$\begin{aligned}
\gamma(0) &= \sigma^2 \sum_{i=0}^{\infty} \psi_i^2 \\
&= \sigma^2 \left\{ 1 + (\phi-\theta)^2 + \sum_{i=2}^{\infty} \phi^{2(i-1)} (\phi-\theta)^2 \right\} \\
&= \sigma^2 \left\{ 1 + (\phi-\theta)^2 \sum_{i=1}^{\infty} \phi^{2(i-1)} \right\} \\
&= \sigma^2 \left(1 + (\phi-\theta)^2 \frac{1}{1-\phi^2} \right),
\end{aligned}$$

and for $l \geq 1$ we have

$$\begin{aligned}
\gamma(l) &= \sigma^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+l} \\
&= \sigma^2 \{ \psi_l + \sum_{i=1}^{\infty} \phi^{i-1} (\phi-\theta)^2 \phi^{i+l-1} \} \\
&= \sigma^2 \left\{ \phi^{l-1}(\phi-\theta) + (\phi-\theta)^2 \sum_{i=1}^{\infty} \phi^{2i+l-2} \right\} \\
&= \sigma^2 \left\{ \phi^{l-1}(\phi-\theta) + \phi^l (\phi-\theta)^2 \frac{1}{1-\phi^2} \right\} \\
&= \sigma^2 \phi^l \left\{ \frac{\phi-\theta}{\phi} + \frac{(\phi-\theta)^2}{1-\phi^2} \right\} \\
&= \sigma^2 (\phi-\theta) \phi^l \left\{ \frac{1}{\phi} + \frac{\phi-\theta}{1-\phi^2} \right\} \\
&= \sigma^2 \phi^l \frac{(\phi-\theta)(1-\phi\theta)}{\phi(1-\phi^2)}.
\end{aligned}$$

That is,

$$\begin{aligned}
\gamma(l) &= \sigma^2\,\phi^l\,k(\phi,\theta), \quad l \geq 1 \\
\gamma(0) &= \sigma^2\,k^*(\phi,\theta)
\end{aligned}$$

for certain functions k and k^*.

Therefore,

$$\rho(l) = c\,(\phi, \theta)\, \phi^l$$

for a certain function $c(\phi, \theta)$. Note that this means that the autocorrelation function eventually decreases like an AR model.

6 Partial autocorrelation in ARMA models

Recall the definition of partial autocorrelation α introduced in Lecture 5.

If X is an AR(1) model we have $\rho_2 = \rho_1^2$ and therefore $\alpha(2) = 0$. In general, for an AR(p) model we have

$$\alpha(k) = 0, \quad \forall k \geq p+1.$$

But if X is an MA(1) we have $\rho_2 = 0$. Then,

$$\alpha(2) = -\frac{\rho_1^2}{1-\rho_1^2}$$

and using

$$\rho_1 = \alpha_1 = -\frac{\theta}{1+\theta^2}$$

we obtain

$$\alpha(2) = \frac{-\theta^2/(1+\theta^2)^2}{1-\theta^2/(1+\theta^2)^2} = \frac{-\theta^2}{(1+\theta^2)^2-\theta^2} = \frac{-\theta^2}{1+\theta^4+\theta^2}.$$

For an AR(p) model, the partial autocorrelation is null for any lag greater than p. On the contrary, for an MA model, the partial autocorrelation decreases exponentially.

7 Non centered ARMA models

It is said that a process $\{Y_j,\ j \in \mathbb{Z}\}$ is an ARMA process with mean μ if

$$X_j := Y_j - \mu, \quad j \in \mathbb{Z}, \tag{7.4}$$

is an ARMA process.

For example in the case $p = q = 1$ we have

$$X_j - \phi\, X_{j-1} = Z_j - \theta\, Z_{j-1}$$

implies

$$(Y_j - \mu) - \phi(Y_{j-1} - \mu) = Z_j - \theta\, Z_{j-1}$$

or equivalently

$$Y_j - \phi\, Y_{j-1} = (\mu - \phi\mu) + Z_j - \theta\, Z_{j-1} = \delta + Z_j - \theta\, Z_{j-1},$$

with

$$\delta = \mu(1 - \phi).$$

Notice also that from (**??**) we have immediately

$$Y_j = \mu + \sum_{i=0}^{\infty} \psi_i\, Z_{j-i}.$$

8 List of exercises

1. Determine what of the following series are causal or invertible. Recall that Z is a white noise.
   (a) $Y_j + 0.2Y_{j-1} - 0.48Y_{j-2} = Z_j$.
   (b) $Y_j + 1.9Y_{j-1} - 0.88Y_{j-2} = Z_j + 0.2Z_{j-1} + 0.7Z_{j-2}$.
   (c) $Y_j + 0.6Y_{j-2} = Z_j + 1.2Z_{j-1}$
   (d) $Y_j + 1.8Y_{j-1} + 0.81Y_{j-2} = Z_j$
   (e) $Y_j + 1.6Y_{j-1} = Z_j - 0.4Z_{j-1} - 0.04Z_{j-2}$.

2. Consider the time series

   $$Y_j = 0.4Y_{j-1} + 0.45Y_{j-2} + Z_j + Z_{j-1} + 0.25Z_{j-2}$$

   where $Z \sim WN(0, \sigma^2)$.
   (a) Write the equation in terms of the lag operator B and show that the models is causal.
   (b) Can this model be simplified? In this case, what are the values (p, q) of the model?
   (c) Find the general form of the coefficients of the causal expression of this series.

3. Consider process $\{X_j, j \in \mathbb{Z}\}$ such that $X_j = \phi X_{j-2} + Z_j$ where $|\phi| < 1$ and Z is a standard white noise. Find the auto-covariance function, the auto-correlation function and the partial auto-correlation function and identify the model.

4. Let $\{X_j, j \in \mathbb{Z}\}$ be a process such that $X_j = Z_j + \theta Z_{j-2}$ where Z is a standard white noise.
   (a) Find auto-covariance, auto-correlation and partial auto-correlation functions for $k = 1, 2, 3, 4$.
   (b) Identify the model.
   (c) Compute the variance of $\frac{1}{4}\sum_{j=1}^{4} X_j$ for $\theta = 0.8, 0$ and -0.8. Comment the results.

5. Let $Y_j = X_j + W_j$ be a time series where W a white noise with variance τ^2 and X is a centered AR(1) model with parameters ϕ and σ^2. Assume the white noise associated to X is uncorrelated with W.
   (a) Show the series Y is stationary and find its auto-covariance function.
   (b) Show the series $U_j = Y_j - \phi Y_{j-1}$ is an MA(1) process.
   (c) Show that Y is an ARMA(1,1) process.

References

[1] P. J. Brockwell and R. A. Davis (1991): *Time Series: Theory and Methods*. Springer.

[2] A. I. Markushévich (1981): *Sucesiones recurrentes*. MIR.